

Adán Flores Ramírez

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EDUCATION

Instituto Tecnológico y de Estudios Superiores de Monterrey <i>B.S. Mechatronics Engineering (96/100); Leadership in Professional Development Award</i>	Monterrey, N.L., Mexico Aug. 2021 – Jun. 2025
Linux Foundation Scholarship recipient, PyTorch Conference 2025	San Francisco, CA, USA

WORK EXPERIENCE

Founding Engineer <i>Riko MX</i> Lead 3-person team building AI voice agents for restaurants; \$8K MRR, \$110K raised. Engineered real-time speech pipeline with sub-second latency and 95% order accuracy using Python/LiveKit/FastAPI	July 2025 – Feb. 2026 Monterrey, N.L., Mexico
Research Lead <i>RoBorregos Robotics Team</i> - Led 15-person software team developing autonomous service robot for RoboCup @HOME competition; achieved 6th place internationally. - Created modular ROS2 framework integrating navigation, manipulation, and perception for domestic environments on Nvidia Jetson embedded hardware. - Designed and deployed on-device command interpretation using fine-tuned LLM with custom parsing architecture achieving 96.5% success rate on diverse commands. - Implemented MoveIt trajectory planning with depth camera for collision-aware grasping; YOLOv8 for real-time object detection. - Established CI/CD pipeline with Docker containerization ensuring reproducible deployments across diverse hardware configurations.	Nov. 2022 – July 2025 Monterrey, Mexico
Undergraduate Research Assistant <i>MIT – Device Realization Lab</i> - Designed ML pipeline analyzing operator behavior in immersive manufacturing environments; simulation experiments demonstrated 29% throughput improvement. - Built data collection infrastructure and conducted user studies for VR-based training systems. - Refactored FrED Factory Python codebase: restructured into modular classes, improved OpenCV optimizations and multithreading, established as standard repository for lab researchers.	Sep. 2024 – Dec. 2024 Cambridge, MA
Software Engineer Intern <i>Google – Cloud AI & Industry Solutions</i> - Built data synchronization components for Vertex AI platform serving enterprise ML workloads. - Reduced computational resources by 40% through C++ algorithm optimization; designed comprehensive testing infrastructure ensuring reliability during major platform migration.	June 2024 – Sep. 2024 Sunnyvale, CA

PROJECTS & RESEARCH

IEEE/ICRCA 2026 <i>Software Toolkit for RoboCup@Home</i> 1st author. Modular hierarchical software architecture for service robots with standardized ROS2 interfaces, deterministic state machines for task management, and containerized multi-platform deployment. Validated on physical robot in competition environment. Accepted. <i>Technologies: ROS2, Docker, MoveIt, YOLO, Python</i>	Jun. – Aug. 2025
MICAI 2025 (Springer LNAI) <i>Natural Language Processing, LLMs, LH Task Planning</i> 1st author: <i>Taming the LLM: Reliable Task Planning for Robotics using Parsing and Grounding</i>. Open-source fine-tuned 8B model achieved 96.5% success vs. cloud APIs (90.4%) in long-horizon task planning for robotics. 2nd Best Student Paper Award. Published.	June - Nov. 2025

TECHNICAL SKILLS

Robotics: ROS/ROS2, MoveIt, Navigation2, Gazebo, SLAM, Sensor Fusion
Languages: Python, C++, Bash, SQL, Go
ML/Vision: PyTorch, YOLO, OpenCV, Transformers, LLMs
Systems: Docker, Linux, Nvidia Jetson, Git, CI/CD